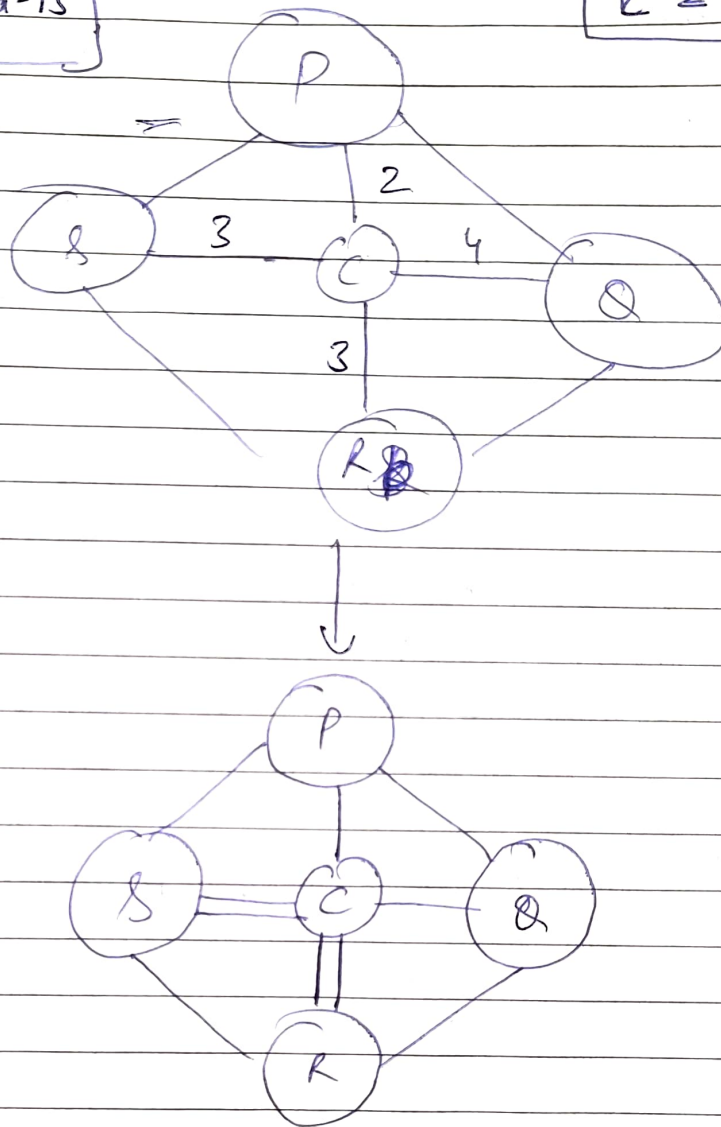


~~scribble~~

Date \_\_\_\_\_  
Page \_\_\_\_\_

github: arshit-mishra-15

E = 122438



First guess → Just like the original problem that Euler solved, this too would be impossible to traverse without repeating.

My path →  $P \rightarrow S \rightarrow R \rightarrow Q \rightarrow C \rightarrow S \rightarrow C \rightarrow R \rightarrow C \rightarrow P \rightarrow Q \rightarrow Q$

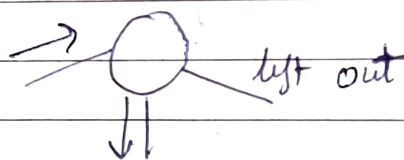
All path traversed !!

This was the first attempt I tried.

| Node | Order | Parity |
|------|-------|--------|
| P    | 3     | odd    |
| Q    | 3     | odd    |
| R    | 4     | Even   |
| S    | 4     | Even   |
| C    | 6     | Even   |

$\therefore$  The number of odd nodes is 2, and the rest are even, the traversal with repeating is possible.

Logic - If a node has odd order, it can be start or end, -



$\therefore$  If there are more than 2 odd order nodes (one start & one end), it cannot be traversed.

### Bonus -

- Method to finding walk that would work on any map.

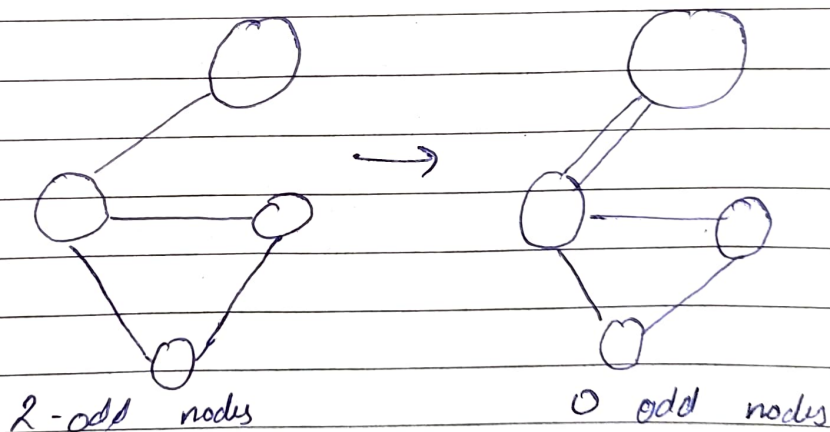
Start with odd degree node if exists, and make the other odd degree as end, and traverse the map.

If a node has 0 odd degree nodes, then it is always possible to traverse. Conversely, if it's more than 2, then it is impossible.

- Euler said no to the great Königsberg because it had 4 - odd parity order bridges (nodes). One can be made as start & one as end, but the rest would be left and entry / exit uses a pair, hence one would always be left out.
- Since the odd ~~per~~ order of bridges must exist in pair, and for the traversal to occur the number of odd degree must be 2 or less -

$$N \geq 4$$

for a given graph to be ~~no~~ non-traversable; when  $N$  is the odd order Nodes.



⇒ Adding one edge ~~with~~ min. converts two nodes

The total  $n$  no. of Nodes needs to be removed is  $(N-2)$ ; we leave 2 nodes for starting and ending.

Adding each bridge ~~at~~ removes two nodes.

⇒  $\left(\frac{N-2}{2}\right)$  would be the min no. of bridges needs to be added

## What surprised me -

as) & My bridge being traversable.

## So I think what's happening is that we need to first convert the bridges to vertices and lands to nodes.

## The pattern being covered here is mainly the odd-even parity.

## Did I use AI? For what?

as) Yes, to write this very last section. Not the content itself but rather how to write.